Examiner only

9. *(a)* A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

	A	**B**	**C**
Add dilute HCl	no reaction	fizzes	no reaction
Add $BaCl_2(aq)$	white precipitate forms	no reaction	no reaction
Add NaOH(aq)	green precipitate forms	pungent smelling gas given off, turns damp red litmus paper blue	no reaction
Add $AgNO_3(aq)$	no reaction	no reaction	cream precipitate forms
Flame test	no colour	no colour	apple-green flame

Use the information provided to give the **chemical name** for each of the compounds. [3]

Compound **A** ....

Compound **B** ....

Compound **C** ....

Examiner only

(b) A technician wants to prepare 250 cm^3 of a 0.25 mol / dm^3 solution of lead nitrate, $Pb(NO_3)_2$.

(i) Calculate the number of moles of lead nitrate required to make the solution. [2]

Number of moles = ... mol

(ii) Calculate the mass of solid lead nitrate that should be dissolved to make the solution. [2]

A_r(Pb) = 207 A_r(O) = 16 A_r(N) = 14

Mass = ... g

(iii) The only electronic balance available to the technician has a precision of ±0.01 g.

Exactly how much lead nitrate should the technician weigh out to ensure that the concentration of the solution is as close as possible to 0.25 mol / dm^3? [1]

Mass .... g

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END OF PAPER